Problem 1/20

GROUP BY

AMAZON

High Sales

Find categories with total sales exceeding $1000.

SAMPLE INPUT

Cat	Amt
A	600
A	500
B	200

EXPECTED OUTPUT

Cat	Total
A	1100

SOLUTION

MYSQL / SQL SERVER

```sql
-- Group by Category, filter Sum
SELECT
    Cat,
    SUM(Amt) AS Total
FROM Sales
GROUP BY Cat
HAVING SUM(Amt) > 1000;
```

Next: Window Functions →

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Problem 2/20

DENSE_RANK

MICROSOFT

2nd Highest

Find the employee with the 2nd highest salary.

SAMPLE INPUT

Emp	Sal
A	100
B	200
C	300

EXPECTED OUTPUT

Emp	Sal
B	200

SOLUTION

MYSQL / SQL SERVER

```sql
-- Rank Rows, then filter 2
WITH Ranked AS (
  SELECT *,
  DENSE_RANK() OVER(ORDER BY Sal DESC) as rnk
  FROM Employees
)
SELECT Emp, Sal
FROM Ranked
WHERE rnk = 2;
```

Next: Null Handling →

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Problem 3/20

COALESCE

GOOGLE

Fill Nulls

Return the first available contact (Phone or Email).

SAMPLE INPUT

ID	Ph	Em
1	NULL	a@x.com
2	123	NULL

EXPECTED OUTPUT

ID	Contact
1	a@x.com
2	123

SOLUTION

MYSQL / SQL SERVER

```sql
-- Returns first non-null value
SELECT
  ID,
  COALESCE(Ph, Em, 'None') as Contact
FROM Users;
```

Next: String Agg →

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Problem 4/20

CASE WHEN

AIRBNB

Price Label

Categorize listings into Budget or Premium.

SAMPLE INPUT

ID	Price
1	50
2	200

EXPECTED OUTPUT

ID	Label
1	Budget
2	Premium

SOLUTION

MYSQL / SQL SERVER

```sql
-- Conditional Logic
SELECT ID,
  CASE
    WHEN Price < 100 THEN 'Budget'
    ELSE 'Premium'
  END AS Label
FROM Listings;
```

Next: Year over Year →

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The 20 Pattern SQL Cheatsheet

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REPOST

Problem 5/20

LAG

META

YoY Growth

Calculate Year-over-Year revenue growth.

SAMPLE INPUT

Yr	Rev
2023	100
2024	120

EXPECTED OUTPUT

Yr	Growth
2023	NULL
2024	20

SOLUTION

MYSQL / SQL SERVER

```sql
-- (Curr - Prev)
WITH Prev AS (
  SELECT Yr, Rev,
  LAG(Rev) OVER(ORDER BY Yr) as LastYr
  FROM Revenue
)
SELECT Yr, (Rev - LastYr) as Growth
FROM Prev;
```

Next: Self Join →

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SAVE

REPOST

Problem 6/20

SELF JOIN

UBER

Manager Sal.

Find employees earning more than their manager.

SAMPLE INPUT

Emp	Sal	Mgr
A	90	B
B	80	NULL

EXPECTED OUTPUT

Emp
A

SOLUTION

MYSQL / SQL SERVER

```sql
-- Join table to itself on Mgr ID
SELECT E.Emp
FROM Staff E
JOIN Staff M ON E.Mgr = M.Emp
WHERE E.Sal > M.Sal;
```

Next: Running Totals →

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The 20 Pattern SQL Cheatsheet

SAVE

REPOST

Problem 7/20

SUM OVER

STRIPE

Running Total

Calculate cumulative revenue day by day.

SAMPLE INPUT

Day	Rev
1	10
2	20
3	30

EXPECTED OUTPUT

Day	Cumul
1	10
2	30
3	60

SOLUTION

MYSQL / SQL SERVER

```sql
-- Unbounded Preceding Sum
SELECT Day, Rev,
  SUM(Rev) OVER (ORDER BY Day) as Cumul
FROM Sales;
```

Next: String Ops →

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The 20 Pattern SQL Cheatsheet

SAVE

REPOST

Problem 8/20

SUBSTRING

TIKTOK

Get Domain

Extract the domain part from email addresses.

SAMPLE INPUT

Email
user@x.com
dev@y.org

EXPECTED OUTPUT

Domain
x.com
y.org

SOLUTION

MYSQL / SQL SERVER

```sql
-- Find @ index, take rest
SELECT
 SUBSTRING(Email, CHARINDEX('@', Email)+1, LEN(Email))
 AS Domain
FROM Users;
```

Next: Advanced Filtering →

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The 20 Pattern SQL Cheatsheet

SAVE

REPOST

Problem 9/20

EXISTS

ORACLE

Active Users

Find users who have placed at least one order.

SAMPLE INPUT

User
A
B

EXPECTED OUTPUT

User
A

SOLUTION

MYSQL / SQL SERVER

```
-- Efficient check for existence
SELECT User
FROM Users u
WHERE EXISTS (
  SELECT 1 FROM Orders o
  WHERE o.User = u.User
);
```

Next: Pivot Data →

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The 20 Pattern SQL Cheatsheet

SAVE

REPOST

Problem 10/20

PIVOT (CASE)

APPLE

Row to Col

Pivot monthly revenue into columns.

SAMPLE INPUT

Mth	Rev
Jan	10
Feb	20

EXPECTED OUTPUT

Jan	Feb
10	20

SOLUTION

MYSQL / SQL SERVER

```sql
-- Manual Pivot using Case
SELECT
  SUM(CASE WHEN Mth='Jan' THEN Rev END) as Jan,
  SUM(CASE WHEN Mth='Feb' THEN Rev END) as Feb
FROM Sales;
```

Next: Distinct Count →

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The 20 Pattern SQL Cheatsheet

SAVE

REPOST

Problem 11/20

COUNT DISTINCT

SHOPIFY

Unique Itms

Count number of unique products sold per day.

SAMPLE INPUT

Day	Prod
1	A
1	A
1	B

EXPECTED OUTPUT

Day	Unique
1	2

SOLUTION

MYSQL / SQL SERVER

```sql
-- Count unique values only
SELECT Day,
  COUNT(DISTINCT Prod) as Unique
FROM Sales
GROUP BY Day;
```

Next: Recursive Logic →

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The 20 Pattern SQL Cheatsheet

SAVE

REPOST

Problem 12/20

RECURSIVE CTE

NETFLIX

Generate Seq

Generate a sequence of numbers from 1 to 3.

SAMPLE INPUT

Start	End
1	3

EXPECTED OUTPUT

Num
1
2
3

SOLUTION

MYSQL / SQL SERVER

```sql
-- Recursive Loop
WITH CTE AS (
  SELECT 1 as Num
  UNION ALL
  SELECT Num + 1 FROM CTE WHERE Num < 3
)
SELECT Num FROM CTE;
```

Next: Date Math →

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The 20 Pattern SQL Cheatsheet

SAVE

REPOST

Problem 13/20

DATE MATH

DOORDASH

Days Diff

Calculate days between Order and Delivery.

SAMPLE INPUT

Ord	Del
2023-01-01	2023-01-05

EXPECTED OUTPUT

Days
4

SOLUTION

SQL SERVER

```sql
-- Date Difference
SELECT
  DATEDIFF(day, Ord, Del) as Days
FROM Logistics;
```

Next: Cross Join →

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The 20 Pattern SQL Cheatsheet

SAVE

REPOST

Problem 14/20

CROSS JOIN

SEPHORA

All Combos

Generate all combinations of Colors and Sizes.

SAMPLE INPUT

Color
Red
Blue

EXPECTED OUTPUT

Color	Size
Red	S
Red	M
Blue	S
Blue	M

SOLUTION

MYSQL / SQL SERVER

```sql
-- Cartesian Product
SELECT C.Color, S.Size
FROM Colors C
CROSS JOIN Sizes S;
```

Next: Union →

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The 20 Pattern SQL Cheatsheet

SAVE

REPOST

Problem 15/20

UNION ALL

TESLA

Append Data

Combine rows from Online and Offline sales (Keep duplicates).

SAMPLE INPUT

Src	Amt
Web	10
Store	20

EXPECTED OUTPUT

Src	Amt
Web	10
Store	20

SOLUTION

MYSQL / SQL SERVER

```sql
-- Stack datasets
SELECT Src, Amt FROM Online
UNION ALL
SELECT Src, Amt FROM Offline;
```

Next: Window Percentile →

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The 20 Pattern SQL Cheatsheet

SAVE

REPOST

Problem 16/20

PERCENTILE

LINKEDIN

Top 50%

Filter for the top 50% of students by score.

SAMPLE INPUT

Stu	Score
A	90
B	50

EXPECTED OUTPUT

Stu	Tile
A	1
B	2

SOLUTION

MYSQL / SQL SERVER

```sql
-- Split into 2 buckets
SELECT Stu,
  NTILE(2) OVER(ORDER BY Score DESC) as Tile
FROM Exams;
```

Next: Left Join →

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The 20 Pattern SQL Cheatsheet

SAVE

REPOST

Problem 17/20

LEFT JOIN

SPOTIFY

No Plays

Find users who never listened to a song.

SAMPLE INPUT

User	Song
A	Pop
B	NULL

EXPECTED OUTPUT

User
B

SOLUTION

MYSQL / SQL SERVER

```sql
-- Filter for Null on Right side
SELECT U.User
FROM Users U
LEFT JOIN Plays P ON U.ID = P.UserID
WHERE P.Song IS NULL;
```

Next: Lead →

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The 20 Pattern SQL Cheatsheet

SAVE

REPOST

Problem 18/20

LEAD

COINBASE

Next Value

Fetch the next day's stock price on the current row.

SAMPLE INPUT

Day	Price
1	100
2	105

EXPECTED OUTPUT

Day	Next
1	105
2	NULL

SOLUTION

MYSQL / SQL SERVER

```sql
-- Look ahead 1 row
SELECT Day,
  LEAD(Price) OVER(ORDER BY Day) as Next
FROM Stocks;
```

Next: Filter Groups →

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The 20 Pattern SQL Cheatsheet

SAVE

REPOST

Problem 19/20

HAVING

WALMART

Group Filter

Find stores with more than 1 employee.

SAMPLE INPUT

Store	Emp
A	John
A	Doe
B	Smith

EXPECTED OUTPUT

Store	Cnt
A	2

SOLUTION

MYSQL / SQL SERVER

```sql
-- Filter after aggregation
SELECT Store, COUNT(*) as Cnt
FROM Staff
GROUP BY Store
HAVING COUNT(*) > 1;
```

Next: De-Dupe →

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The 20 Pattern SQL Cheatsheet

SAVE

REPOST

Problem 20/20

ROW_NUMBER

TINDER

De-Dupe

> Keep only the most recent login per user.

SAMPLE INPUT

User	Date
A	Jan1
A	Jan5

EXPECTED OUTPUT

User	Date
A	Jan5

SOLUTION

MYSQL / SQL SERVER

```sql
-- Number rows per user desc
SELECT User, Date FROM (
  SELECT *,
  ROW_NUMBER() OVER(PARTITION BY User ORDER BY Date DESC)
rn
  FROM Logins
) t WHERE rn = 1;
```

Next: MISSION ACCOMPLISHED →

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The 20 Pattern SQL Cheatsheet

SAVE

REPOST

MISSION ACCOMPLISHED

You are Interview Ready!

20 PATTERNS MASTERED 100%

- [x] GROUP BY
- [x] DENSE_RANK
- [x] COALESCE
- [x] CASE WHEN
- [x] LAG
- [x] SELF JOIN
- [x] SUM OVER
- [x] SUBSTRING
- [x] EXISTS
- [x] PIVOT (CASE)
- [x] COUNT DISTINCT
- [x] RECURSIVE CTE
- [x] DATE MATH
- [x] CROSS JOIN
- [x] UNION ALL
- [x] PERCENTILE
- [x] LEFT JOIN
- [x] LEAD
- [x] HAVING
- [x] ROW_NUMBER

FOLLOW FOR MORE ♥

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